

Manthan Bhala

Machine Learning Engineer & Software Developer

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EXPERIENCE

Senior Machine Learning Engineer — Makro PRO

Oct 2023 – Present

Remote

- Built recommendation models using collaborative filtering, two-tower networks, and vector embeddings.
- Increased total basket additions by 52% through personalised user-based and product-based recommendations.
- Built vector search by fine-tuning LLM embeddings for semantic product retrieval.
- Built personalisation via e-commerce feature engineering and LLM distillation into LightGBM.
- Developed end-to-end recommendation pipeline with retrieval, ranking, and reranking stages.
- Improved search with autocategorization, autocorrect, query expansion, and query rewriting.
- Built a chatbot POC with RAG, function calling, quantization, and vLLM deployment.

Machine Learning Engineer — Frinks AI

Mar 2023 – Jul 2023

Bengaluru, Karnataka, India

- Developed ML inference pipelines for real-time defect detection on production lines.
- Built CV and backend services for quality inspection: dimensioning, defect, and crack detection.

Data Scientist — Jaguar Land Rover

Sep 2020 – Feb 2023

Bengaluru, Karnataka, India

- Developed ML pipelines and backend services for Vehicle Algorithmic Self Diagnostics Tool.
- Worked on ADAS systems for autonomous and assisted driving features.

SKILLS

ML Engineering: Python · PyTorch · TensorFlow · Hugging Face · LightGBM · PEFT

Applied AI: LLMs · Recommendation Systems · Information Retrieval · Learning to Rank · RAG · Vector Search · Computer Vision · NLP

Data & Infrastructure: SQL · Docker · Kubernetes · CI/CD · Redis · vLLM

EDUCATION

Bachelor of Technology — Indian Institute of Technology (IIT), Bombay

2016 – 2020 | CGPA: 8.48/10

PATENTS

2501007631 — A Hybrid Lexical-Semantic Retrieval Process for E-Commerce Search

2501004476 — A Hybrid LLM Process for Multi-Objective Reranking in E-Commerce

2501004419 — A Multilingual Phonetic BK-Tree Based Spelling Correction Process in E-Commerce